Data Collection

Solutions

This Chapter Overview

Interestingly, most of this chapter is from the old S3 module (a Further Maths module!) with also some S2. There is little 'calculation' involved in this chapter; consider this a 'bookwork' one!

1: Populations vs samples

"Suggest why we would not test all the light bulbs."
"Identify the sampling frame."

2: Random Sampling

Describe the disadvantages of systematic sampling.

3: Non-Random Sampling

Describe how a stratified sample would be conducted, including strata sizes.

4: Types of data

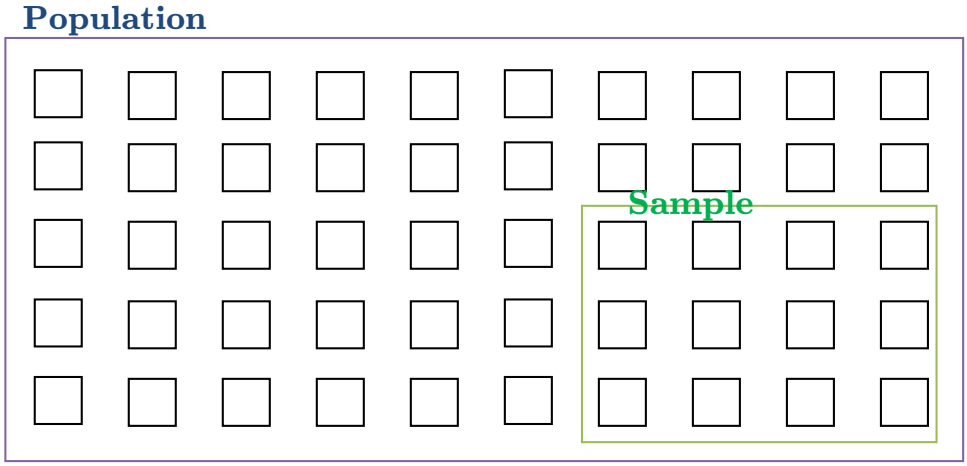
Continuous vs discrete, terms such as class intervals, class boundaries, class width.

5: Edexcel's 'Large Data Set'

What you're expected to know about the 'large data set' of weather data, and how to use it.

Populations and samples

Solution 3



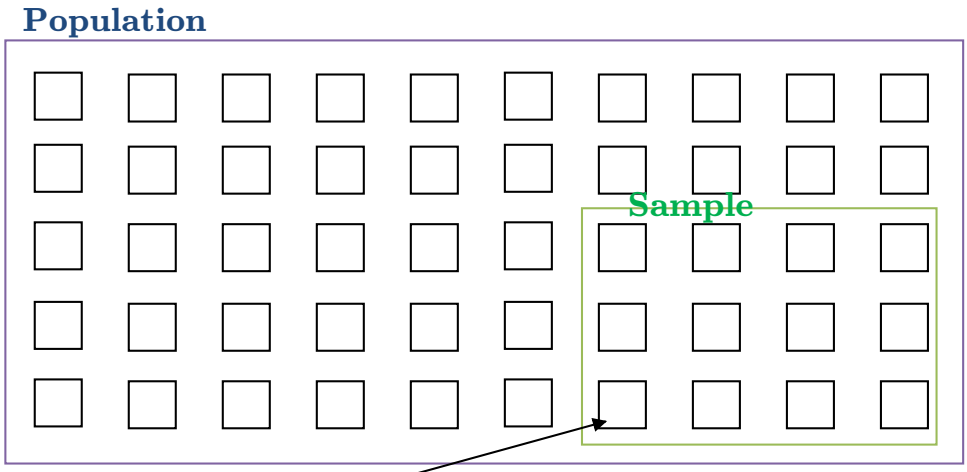
A population is the whole set of items that are of interest.

A sample is a subset intended to represent the population.

You're probably used to a 'population' meaning all humans/animals within a country/ecosystem. But a population could be "*all the lightbulbs in a factory*" or "*all the cars in the UK*".

Sampling key terms

Solution 4



Each individual thing in the population that can be sampled is known as a **sampling unit**.

Often sampling units of a population are individually named or numbered **to form a list** called the **sampling frame**.

Populations vs Samples

Solution 5

We could collect data either from a sample, or from the entire population.
Data collected from the entire population is known as a census.

	Advantages	Disadvantages
Census	Should give completely accurate result.	• Time consuming and expensive. • Can not be used when testing involves destruction. • Large volume of data to process.
Sample	• Cheaper. • Quicker. • Less data to process.	• Data may not be accurate. • Data may not be large enough to represent small sub-groups.

Example: A supermarket wants to test a delivery of avocados for ripeness by cutting them in half.

a. Suggest a reason why the supermarket should not test all the avocados in the delivery.

b. The supermarket tests a sample of 5 avocados and finds that 4 of them are ripe. They estimate that 80% of the avocados in the deliver are ripe. Suggest one way that the supermarket could improve their estimate.

a

a Testing the avocados destroys them (and thus can't be sold).

b

b Use a larger sample size (as this would be better estimate of the proportion of ripe avocados).

Exercise 1A – Pearson Statistics Y1 page 3

Solution 6

1. A school uses a census to investigate the dietary requirements of its students.

(a) Explain what is meant by a census.

(b) Give one advantage and one disadvantage to the school of using a census.

a

A census observes or measures every member of a population.

b

It gives a completely accurate result.
A disadvantage could be either of the following.
It would be time consuming.
It would be expensive.

Exercise 1A – Pearson Statistics Y1 page 3 — Solution 7

2. A factory makes safety harnesses for climbers and has an order to supply 3000 harnesses. The buyer wishes to know that the load at which the harness breaks exceeds a certain figure.

(a) Suggest a reason why a census would not be used for this purpose.

The factory tests four harnesses and records their breaking loads.

320 kg, 260 kg, 240 kg, 180 kg

(b) The factory claims that the harnesses are safe for loads up to 250 kg. Use the sample data to comment on this claim.

(c) Suggest one way in which the company can improve its prediction.

a

Testing destroys the harness. A census would destroy every harness, leaving none for climbers to use.

b

The claim is misleading. The mean and median breaking loads are both 250 kg, and half the sample broke below 250 kg.

c

Test a larger number of harnesses.

Exercise 1A – Pearson Statistics Y1 page 3 — Solution 8

3. A city council wants to know what people think about its recycling centre. It decides to carry out a sample survey of residents.

(a) Write down one reason why the council should not take a census.

(b) Suggest a suitable sampling frame.

(c) Identify the sampling units.

3a

Any one of the following. It would take too long, it would be expensive, or the data would be difficult to process.

3b

A list of residents.

3c

Each individual resident.

5. A garage manager wants to know what their mechanics think about a new pension scheme designed for them. The manager asks every mechanic in the garage.

(a) Describe the population the manager will use.

(b) Write down the main advantage of asking all the mechanics.

5a

All the mechanics in the garage.

5b

Every mechanic's view will be known.

Types of Sampling

Solution 9

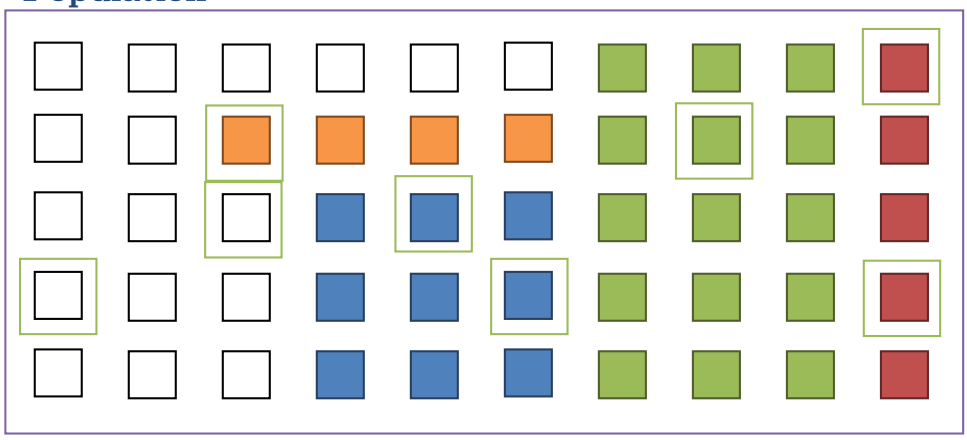
Lay out your notes like this for next bit of the chapter. Use a full page.

	Type	How to carry out	Advantages	Disadvantages
Random Sampling	Simple Random Sampling			
	Systematic Sampling			
	Stratified Sampling			
Non-Random	Quota Sampling			
	Opportunity Sampling			

Random Sampling

Solution 10

Population



Ordinarily, we would want each thing in our sampling frame to have an **equal chance of being chosen**, in order to **avoid bias**.

This is known as **random sampling**.
There are a few ways of doing this. . .

Simple Random Sampling

Solution 11

Type	How to carry out	Advantages	Disadvantages
Simple Random Sampling	**What is it :** Every sample has an equal chance of being selected. **Method:** In sampling frame each item has identifying number. Use random number generator, or 'lottery sampling' (names in a hat).	• Bias free. • Easy and cheap to implement. • Each number has a known equal chance of being selected.	• Not suitable when population size is large. • Sampling frame needed.

Edexcel S3 June 2004 Q1a

There are 64 girls and 56 boys in a school. Explain briefly how you could take a random sample of 15 pupils using a simple random sample. **(3)**

Assign a number from 1 to 120 to each pupil. (M1)
Use a random number table, computer or calculator to select 15 distinct numbers from 1 to 120. (B1)
The corresponding pupils form the sample. (B1)

Mark for allocating an identifier to each sampling unit.

Mark for one bias-free method to select such a number.

Mark for explicitly mentioning how that number is actually used.

Systematic Sampling

Solution 12

Type	How to carry out	Advantages	Disadvantages
Systematic Sampling	**What is it :** Required elements are chosen at regular intervals in ordered list. i.e. Take every k^{th} elements where: $k = \frac{popsize(N)}{sampsize(n)}$ starting at random item between 1 and k.	• Simple and quick to use. • Suitable for large samples/ populations.	• Sampling frame again needed. • Can introduce bias if sampling frame not random.

Edexcel S3 June 2009 Q1a

A telephone directory contains 50 000 names. A researcher wishes to select a systematic sample of 100 names from the directory. Explain in detail how the researcher should obtain such a sample. **(2)**

We need a random first item.

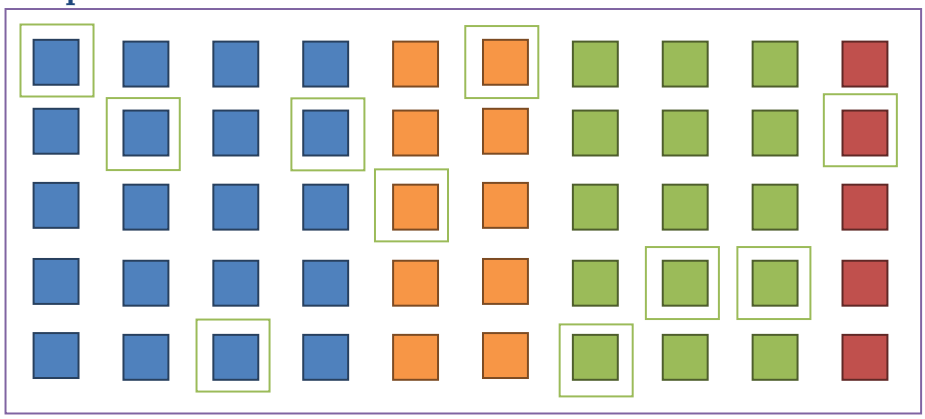
Choose at random one of the first 500 names, then select every 500th person. (B1, B1)

Stratified Sampling

Solution 13

We want to sample 20% of the population. If the population were divided into distinct groups (e.g. age ranges), known as '**strata**', we could randomly sample 20% from each group, ensuring each group is equally represented.

Population



Type	How to carry out	Advantages	Disadvantages
Stratified Sampling	What is it: Population divided into groups (strata), with a simple random sample in each group. Sample the same proportion n/N from each stratum. Used when the sample is large and the population naturally divides into groups.	• Reflects population structure. • Guarantees proportional representation of groups within population.	• Population must be clearly classified into distinct strata. • Selection within each stratum suffers from same disadvantages as simple random sampling.

Example Question

Solution 14

Edexcel S3 Jan 2006 Q1

A school has 15 classes and a sixth form. In each class there are 30 students. In the sixth form there are 150 students. There are equal numbers of boys and girls in each class. There are equal numbers of boys and girls in the sixth form. The head teacher wishes to obtain the opinions of the students about school uniforms. Explain how the head teacher would take a stratified sample of size 40. **(7)**

Total: $15 \times 30 + 150 = 600$. (B1)
Per class: $\frac{30}{600} \times 40 = 2$. Sixth form: $\frac{150}{600} \times 40 = 10$. (M1, A1, A1)
Number boys and girls separately from 1 to 15 in each class. Choose one of each using random numbers. (B1, B1)
In the sixth form number each group from 1 to 75. Choose five distinct boys and five distinct girls using random numbers. (B1)

You would certainly want to know your mark scheme on this one!

Exercise 1B – Pearson Statistics Y1 page 6-7

Solution 15

2. A survey is carried out on 100 members of the adult population of a city suburb. The suburb has 2000 residents. An alphabetical list of the inhabitants is available.

(a) Explain one limitation of using a systematic sample.

(b) Describe a sampling method that would be free of bias for this survey.

2a

Taking every 20th person may introduce bias because the sampling frame is in alphabetical order.

2b

Take a simple random sample using the alphabetical list as the sampling frame.

3. A gym wants to take a sample of its members. Each member has a five-digit membership number. The gym selects every member whose number ends in 000.

(a) Is this a systematic sample? Give a reason.

(b) Suggest one way to improve the reliability of this sample.

3a

No. The first person is not selected at random, and the method does not select elements at regular intervals in the membership list.

3b

Use a larger sample to improve reliability. To reduce bias, take a simple random sample using the membership list as the sampling frame.

Exercise 1B – Pearson Statistics Y1 page 6-7

Solution 16

4. A head of sixth form wants the opinions of Year 12 and Year 13 students about the common room facilities. The table shows the numbers in each year.

	Year 12	Year 13
Male	70	50
Female	85	75

(a) Suggest a suitable sampling method for a sample of 40 students.

(b) How many students from each gender in each year should be asked?

4a Stratified sampling.

4b

$70 + 50 + 85 + 75 = 280$ students. Round each allocation to the nearest whole number.

Year 12 male: $\frac{70}{280} \times 40 = 10$

Year 13 male: $\frac{50}{280} \times 40 \approx 7$

Year 12 female: $\frac{85}{280} \times 40 \approx 12$

Year 13 female: $\frac{75}{280} \times 40 \approx 11$

5. A factory manager wants to find out how workers travel to work. Each of the 480 workers has a clocking-in number from 1 to 480. Explain how to take a systematic sample of size 30.

5

$k = \frac{480}{30} = 16.$

Choose a starting number at random from 1 to 16. Select the worker with that clocking-in number, then take every 16th number after it.

Exercise 1B – Pearson Statistics Y1 page 6-7 Solution 17

6. A sports club director wants a sample of members. Each member has a unique membership number. There are 121 cricket players, 145 hockey players and 104 squash players. No one plays more than one sport.

(a) Explain how to take a simple random sample of 30 and give one disadvantage.

The director decides to take a stratified sample of 30.

(b) State one advantage of this method.

(c) Work out the number of members to select from each sport.

a

Use the membership list as a sampling frame. Draw 30 distinct numbers by lottery, giving every member the same chance of selection. The sample may not reflect the proportions playing each sport.

b Each sport has proportional representation in the sample.

c

Total: $121 + 145 + 104 = 370$.

Cricket: $\frac{121}{370} \times 30 \approx 9.81$.

Hockey: $\frac{145}{370} \times 30 \approx 11.76$.

Squash: $\frac{104}{370} \times 30 \approx 8.43$.

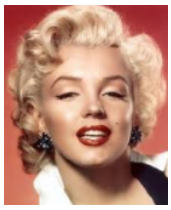
Take 10 cricket, 12 hockey and 8 squash players, giving 30 members.

Non-Random Sampling Solution 18

Consider the following scenario: You wish to conduct a survey in the UK **on whether being left-handed affects IQ**. We need to choose people to assess.

Famous Lefties







Why would random sampling be problematic?

Because we do not know the sampling frame, i.e. do not have a list of all left-handed (and non-left-handed) people in the UK.

For this scenario we'd likely use **quota sampling**, i.e.

1. As with stratified sampling, divide population into groups according to characteristic of interest, then determine size of each group in sample to reflect proportions within the population.
2. But instead of random sampling within each group, we actively choose people within each group via suitable means (e.g. advertising), **until the 'quota' for each group is filled**.

A variant of this is **opportunity sampling**, where we find people **at the same time the survey is being carried out** (e.g. exit polls at polling stations). This is not a suitable method for the left-handed example, because giving the likely time-consuming nature of assessment coupled with resources required, we'd likely arrange with the people taking part before the actual assessment tasks took place.

Quota & Opportunity Sampling

Solution 19

Type	How to carry out	Advantages	Disadvantages
Quota Sampling	**What is it :** Population divided into groups according to characteristic. A quota of items/people in each group is set to try and reflect the group's proportion in the whole population. Interviewer selects the actual sampling units.	• Allows small sample to still be representative of population. • No sampling frame required. • Quick, easy, inexpensive. • Allows for easy comparison between different groups in population.	• Non-random sampling can introduce bias. • Population must be divided into groups, which can be costly or inaccurate. • Increasing scope of study increases number of groups, adding time/expense. • Non-responses are not recorded.
Opportunity/ Convenience Sampling	Sample taken from people who are available at time of study, who meet criteria.	• Easy to carry out. • Inexpensive.	• Unlikely to provide a representative sample. • Highly dependent on individual researcher.

Example Question

Solution 20

Edexcel S3 June 2010 Q2

A lake contains three species of fish. There are estimated to be 1400 trout, 600 bass and 450 pike. A survey of fish health is carried out and a sample of 30 fish is chosen.

(a) Give a reason why stratified random sampling cannot be used. (1)

(b) State an appropriate sampling method for the survey. (1)

(c) Give one advantage and one disadvantage of this method. (2)

(d) Explain how to use this method to select the 30 fish. Show your working. (4)

a A sampling frame within each species is impossible to obtain. (B1)

b Quota sampling. (B1)

c

Advantages (B1): quick, low cost or easy to administer.

Disadvantages (B1): sampling errors cannot be estimated, the process is non-random, or species may be difficult to identify.

d

Species	Quota
Trout	$\frac{1400}{2450} \times 30 = 17.14$
Bass	$\frac{600}{2450} \times 30 = 7.35$
Pike	$\frac{450}{2450} \times 30 = 5.51$

Catch fish until the quotas of 17 trout, 7 bass and 6 pike are reached. (B1 B1 B1)

Ignore a fish if the quota for its species is full. (B1)

Exercise 1C – Pearson Statistics Y1 page 8-9

Solution 21

1. Interviewers collect information about spending habits from 40 shoppers in a shopping centre.
(a) Explain how to use (i) quota sampling and (ii) opportunity sampling.
(b) Which method is likely to give a more representative sample?

1a(i)
Divide the population by characteristic. Set each group's quota in proportion to its size. Identify the group of each interviewee and stop sampling from a group when its quota is full.

1a(ii)
Interview those available at the time, for example the first 40 available shoppers.

1b
Quota sampling. An opportunity sample is unlikely to reflect the characteristics of the whole population.

2. Describe the similarities and differences between quota sampling and stratified random sampling.

2
Both divide the population into groups according to its characteristics. Quota sampling uses non-random selection within groups. Stratified random sampling selects randomly within each stratum.

Exercise 1C – Pearson Statistics Y1 page 8-9

Solution 22

3. An interviewer asks the first 50 people he sees outside a fish and chip shop on a Friday evening about their eating habits.
(a) What sampling method did he use?
(b) Why might the sample be unrepresentative?
(c) Suggest two improvements to data collection.

3a Opportunity sampling.

3b
Customers at this place and time are unlikely to represent the whole population.

3c
Survey people at different times of day and in other parts of town.

4. A researcher asks the first five people entering a supermarket on Monday morning about their weekly radio listening. They report 4, 7, 6, 8 and 2 hours.
(a) Predict the average weekly hours for the town.
(b) Describe the sampling method and comment on reliability.
(c) Suggest two improvements.

4a
$\bar{x} = (4 + 7 + 6 + 8 + 2)/5 = 5.4$ hours.

4b
This is a small opportunity sample. It is unlikely to represent the town.

4c
Increase the sample size. Ask people at other times or in different locations.

Exercise 1C – Pearson Statistics Y1 page 8-9 Solution 23

5. Scientists studying the masses of wild deer catch the first five male and the first five female deer they find.

(a) What sampling method are they using?

(b) Give one advantage.

Male (kg)	75	80	90	85	82
Female (kg)	67	72	75	68	65

(c) Compare the masses of male and female deer.

(d) Suggest two improvements to the sampling method.

a Quota sampling.

b
Any one: no sampling frame is needed, the method is quick, easy or inexpensive, or it allows comparison between males and females.

c
Male mean: 412/5 = 82.4 kg (82 kg to the nearest kg).
Female mean: 347/5 = 69.4 kg (69 kg to the nearest kg).
Ranges: 90 - 75 = 15 kg and 75 - 65 = 10 kg.
Males are heavier on average and their masses have a greater spread.

d Increase the sample size. Catch deer at random times during the day.

Exercise 1C – Pearson Statistics Y1 page 8-9 Solution 24

6. The heights in metres of 20 ostriches are listed below.

1.8, 1.9, 2.3, 1.7, 2.1, 2.0, 2.5, 2.7, 2.5, 2.6, 2.3, 2.2, 2.4, 2.3, 2.2, 2.5, 1.9, 2.0, 2.2, 2.5.

(a) Take an opportunity sample of size five.

(b) Starting at the second value, take a systematic sample of five.

(c) Calculate each sample mean.

(d) State, with reasons, which method is likely to be more reliable.

a For example, the first five heights: 1.8, 1.9, 2.3, 1.7, 2.1.

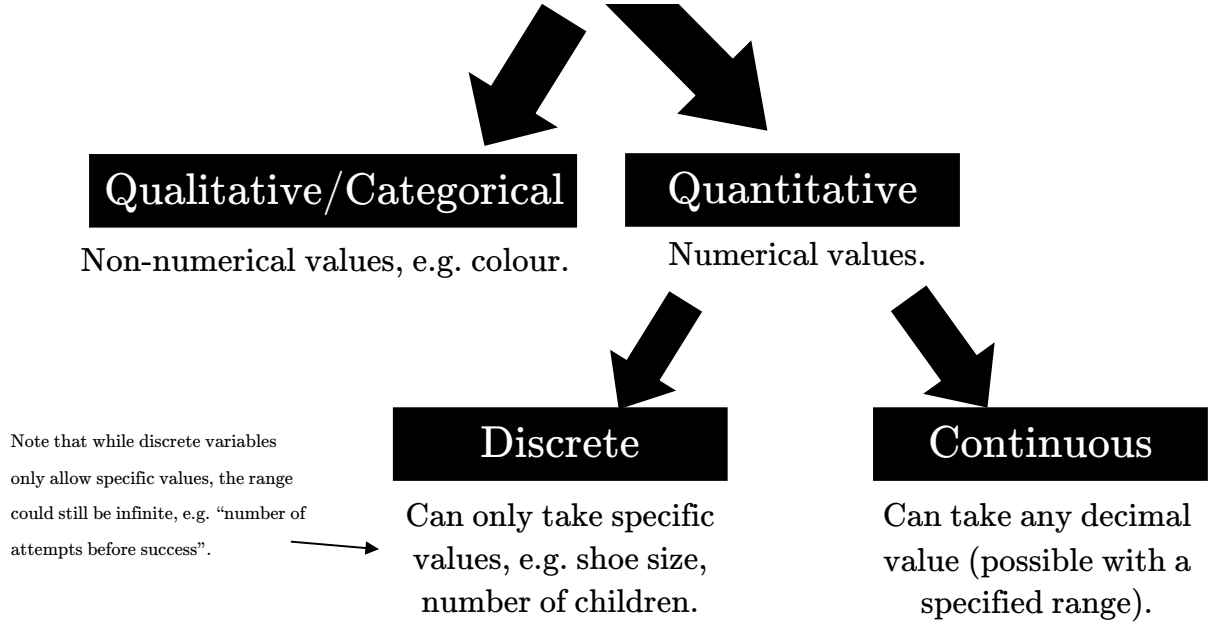
b
Start at 1.9 and take every fourth height: 1.9, 2.0, 2.6, 2.3, 2.0.

c
Opportunity: $\frac{1.8+1.9+2.3+1.7+2.1}{5} = 1.96$ m.
Systematic: $\frac{1.9+2.0+2.6+2.3+2.0}{5} = 2.16$ m.

d
The systematic sample is spread throughout the list and is likely to be more representative. The opportunity sample can include only small values, as it does here.

Types of Data

Solution 25



Weight **w** (kg)	Frequency
$0 \leq w < 20$	3
$20 \leq w < 70$	4

Data can be **grouped** for conciseness, at the expense of losing the exact original values.

$20 \leq w < 70$

This is known as a class interval.

Lower class boundary

Midpoint = 45

Upper class boundary

Class width = $70 - 20 = 50$

Exercise 1D – Pearson Statistics Y1 page 10

Solution 26

1. Is each variable qualitative or quantitative?
(a) Height of a tree
(b) Colour of a car
(c) Time waiting in a queue
(d) Shoe size
(e) Names of pupils in a class

1a Quantitative: height is numerical.

1b Qualitative: colour is a category.

1c Quantitative: time is numerical.

1d Quantitative: shoe size is numerical.

1e Qualitative: names are categories.

2. Is each quantitative variable continuous or discrete?
(a) Shoe size
(b) Length of a leaf
(c) Number of people on a bus
(d) Weight of sugar
(e) Time required to run 100 m
(f) Lifetime in hours of torch batteries

2a Discrete: sizes occur in fixed increments.

2b Continuous: length can be measured to any precision.

2c Discrete: people are counted in whole numbers.

2d Continuous: weight can be measured to any precision.

2e Continuous: time can be measured to any precision.

2f Continuous: lifetime can be measured to any precision.

Exercise 1D – Pearson Statistics Y1 page 10 Solution 27

3. Explain why each statement is true.
(a) Type of tree is qualitative.
(b) Number of pupils in a class is discrete and quantitative.
(c) Weight of a collie dog is continuous and quantitative.

3a Tree type describes a category.

3b The number of pupils is numerical and can only be a whole number.

3c
Weight is numerical and can take any value in a range.

4. The table shows the masses of two-month-old lambs.

Mass m (kg)	Frequency
$1.2 \leq m < 1.3$	8
$1.3 \leq m < 1.4$	28
$1.4 \leq m < 1.5$	32
$1.5 \leq m < 1.6$	22

(a) Give the boundaries of the third group.
(b) Find the midpoint of the second group.
(c) Find the width of the first group.

4a
1.4 and 1.5 kg. There are no gaps between adjacent classes.

4b
$(1.3 + 1.4)/2 = 1.35$ kg.

4c
$1.3 - 1.2 = 0.1$ kg.

Name That Sampling Method! Solution 28

Simple Random Sampling	Systematic Sampling	Stratified Sampling	Quota Sampling	Opportunity Sampling

Suggest a suitable sampling method.

"You wish to test lightbulbs produced by a factory in a daily batch."

Probably **systematic sampling**, as the method of choosing items is simpler than simple random sampling (where it would be time-consuming to find specifically chosen random light bulbs). Sampling frame is known.

"You wish to survey consumer opinion on your new drink *FizzGuzz* released in the UK."

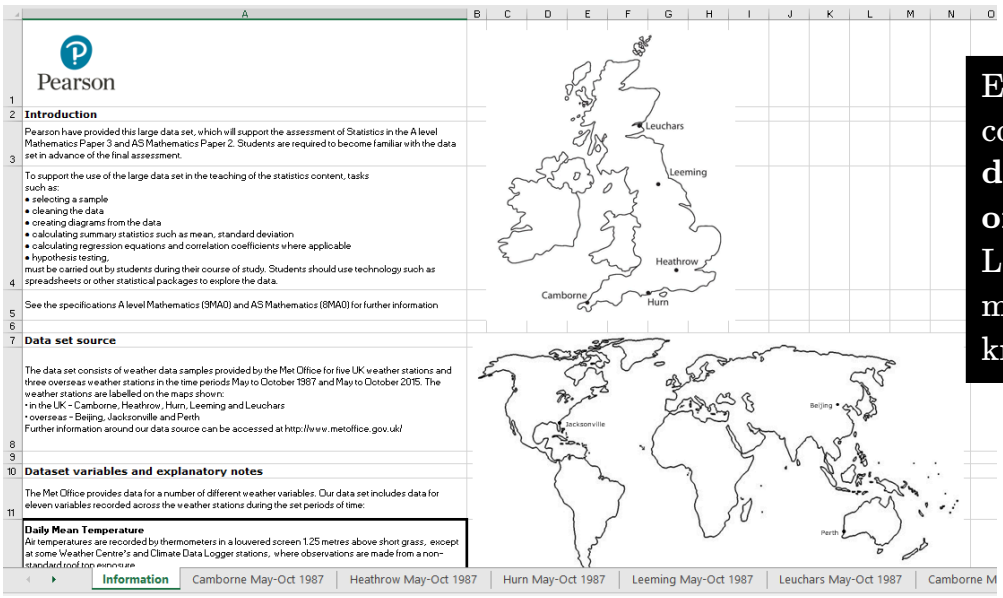
Quota sampling or **opportunity sampling**. We'd realistically not have access to the sampling frame (i.e. a list of all UK residents).

"You wish to determine students' favourite TV programmes in your school, that is fairly representative of each year group."

Stratified sampling. We (probably) have access to the sampling frame (i.e. a list of all students). Stratified sampling ensures that each stratum (year group) is proportionately represented.

All A Level exam boards are obligated to provide a 'large data set'. Data in exam questions will often be from this set, and you are encouraged to explore this data (which is publicly available) in Microsoft Excel.

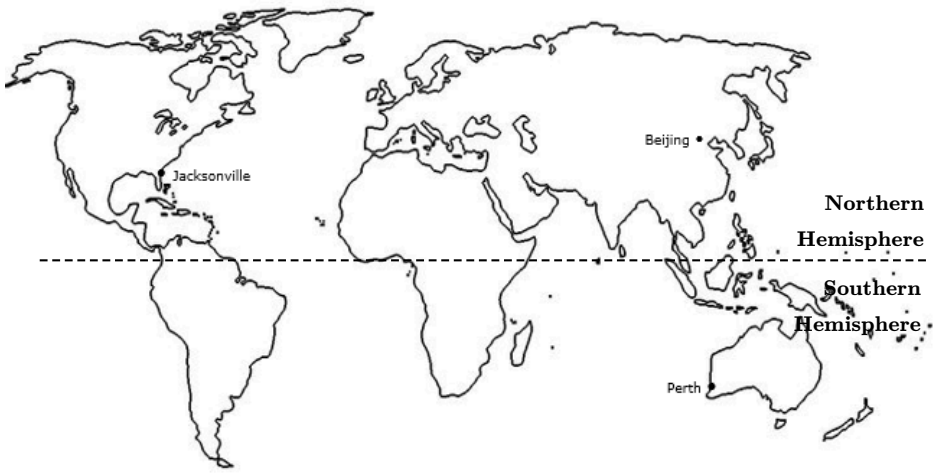
It is important to note that you are expected to be familiar with this data set before you go into your exam, including some basic geographic knowledge!



Edexcel's data set concerns **weather data from a number of weather stations**. Let's explore what you might be expected to know. . .

https://qualifications.pearson.com/content/dam/pdf/A%20Level/Mathematics/2017/specification-and-sample-assesment/Pearson%20Edexcel%20GCE%20AS%20and%20AL%20Mathematics%20data%20set%20-%20Issue%201%20(1).xls





1 You should know the names and rough locations of the 5 UK weather stations, as well as the 3 international weather stations.

The data was recorded for:

- May-Oct 1987
- May-Oct 2015

2 You should be familiar with the variables involved and their respective units.

Total rainfall

(in mm)

tr/trace means less than 0.05mm

Mean Windspeed

kn/knot is "nautical mile per hour". $1kn = 1.15mph$

Windspeed also given on **Beaufort Scale**:

0 = Calm	$< 1kn$
1-3 = Light	1-10kn
4 = Moderate	11-16kn
5 = Fresh	17-21kn

Mean Visibility

How far (in metres) can be seen into the horizon during daylight hours.

Wind Direction

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	CAMBORNE		© Crown Copyright Met Office 2015										
2	NGR = 1627E 4067N												
3	Altitude = 87 metres												
4	Latitude = 50:22N Longitude = 05:33W												
5													
6	Date	Daily Mean Temperature (0900-0900) (°C)	Daily Total Rainfall (0900-0900) (mm)	Daily Total Sunshine (0000-2400) (hrs)	Daily Mean Windspeed (0000-2400) (kn)	Daily Mean Windspeed (0000-2400) (Beaufort conversion)	Daily Maximum Gust (0000-2400) (kn)	Daily Maximum Relative Humidity %	Daily Mean Total Cloud (oktas)	Daily Mean Visibility (Dm)	Daily Mean Pressure (hPa)	Daily Mean Wind Direction (o)	Cardinal Direction	Daily Max Gust Corresponding Direction (o)	Cardinal Direction
7	01/05/1987	10.7	3.1	n/a	n/a	n/a	n/a	100	7	2000	1018	360	N	20	NNE
8	02/05/1987	8.9	0.1	n/a	n/a	n/a	n/a	91	3	3200	1020	320	NW	330	NNW
9	03/05/1987	8.1	0	n/a	n/a	n/a	n/a	77	5	3600	1029	350	N	350	N
10	04/05/1987	8.2	0	n/a	n/a	n/a	n/a	83	5	4100	1036	350	N	350	N
11	05/05/1987	9.8	0	n/a	n/a	n/a	n/a	86	5	2700	1036	10	N	10	N
12	06/05/1987	9.3	0	n/a	n/a	n/a	n/a	100	1	1000	1033	330	N		
13	07/05/1987	10.9	0	n/a	n/a	n/a	n/a	100	3	600	1031	350			
14	08/05/1987	10.5	tr	n/a	n/a	n/a	n/a	89	1	2400	1025	110			
15	09/05/1987	10.9	0	n/a	n/a	n/a	n/a	95	3	900	1017	360			
16	10/05/1987	9.9	0	n/a	n/a	n/a	n/a	79	4	4100	1018	10			
17	11/05/1987	8.8	6	n/a	n/a	n/a	n/a	95	7	2500	1017	270	W	260	W
18	12/05/1987	10.2	tr	n/a	n/a	n/a	n/a	97	5	2400	1009	310	NW	310	NW
1			2.2	n/a	n/a	n/a	n/a	77	4	4600	1016	340	NNW	340	NNW
2			tr	5.9	16	Moderate	33	95	7	3100	1008	290	WNW	270	W
2			0	12.3	13	Moderate	27	77	4	4500	1012	10	N	10	N
2			tr	11.6	6	Light	16	92	4	3700	1015	290	WNW	290	WNW

Mean Pressure

In hectopascals (hPa)

Mean temperature

(in $^{\circ}C$)

Textbook claims this is **max** temp for UK, but it is **mean** temp for all locations.

Total sunshine

(nearest $\frac{1}{10}$ of an hour)

Maximum Gust

(in kn) is highest instantaneous wind speed.

Humidity

is the % of air saturation with water vapour. 100% is the maximum % water content air can contain.

Mean Cloud Cover

Oktas means the number of $\frac{1}{8}$ths of the sky covered.

3 You should have a vague idea of the range of values for each location.

UK Location (2015)	Temp Range	Wind Speed Range
Camborne	10-20	3-18
Heathrow	8-**29**	3-19
Hurn	6-24	2-19
Leeming	4-23	3-17
Leuchars	4-19	3-23

World Location (2015)	Temp Range	Wind Speed Range
Beijing	8-33	2-9
Jacksonville	15-31	1-12
Perth	8-25	4-14

Beijing temp range relatively large.
Min Jacksonville temp high.
Perth similar to UK.

Mean wind speed in UK across full period was roughly 9 nm. But 4 nm in Beijing (i.e. lower), 5 in Jacksonville (again lower), 8 in Perth (similar to UK).

From new A Level sample assessment materials:

"A meteorologist believes that there is a relationship between the daily mean windspeed, w kn, and the daily mean temperature, t °C. A random sample of 9 consecutive days is taken from past records from a town in the UK in July and the relevant data is given in the table below. . . .

Using the same 9 days, a location from the large data set gave $\bar{t} = \mathbf{27.2}$ and $\overline{w} = \mathbf{3.5}$.

(d) Using your knowledge of the large data set, suggest, giving your reason, the location that gave rise to these statistics."

(Note to teachers: I will not otherwise use SAM questions in these slides. I made one exception here!)

4 You should have a vague idea of the range of values for each variable for the data set as a whole.

Variable	Typical value(s)
Gust (UK only)	8 – 52 nm
Rainfall	0 – 60 mm in UK, but more extreme maximums elsewhere (e.g. 102mm in Perth)
Pressure	988 – 1038 hPa
Wind Speed on Beaufort scale	Max is 'fresh' (5). Most Light or Moderate.
Sunshine (UK only)	0 – 16 hrs
Cloud Cover	0 – 8 ocktas (i.e. full spread)

Hurn

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Daily maximum temperature and rainfall: 09:00–09:00. Sunshine, mean wind and maximum gust: 00:00–24:00.

June 1987	Max °C	Rain mm	Sun h	Wind kn	Beaufort	Gust kn
01/06	15.1	.6	4.5	7	Light	19
02/06	12.5	4.7	0	7	Light	22
03/06	13.8	tr	5.6	11	Moderate	25
04/06	15.5	5.3	7.8	7	Light	17
05/06	13.1	19	.5	10	Light	33
06/06	13.8	0	8.9	19	Fresh	46
07/06	13.2	tr	3.8	11	Moderate	27
08/06	12.9	1	1.7	9	Light	19
09/06	11.2	tr	5.4	6	Light	19
10/06	9.2	1.3	9.7	4	Light	n/a
11/06	12.6	0	12.5	6	Light	18
12/06	10.4	0	11.9	5	Light	n/a
13/06	9.6	0	8.6	5	Light	15
14/06	10.2	0	13.1	5	Light	18
15/06	9.2	3.7	7.1	4	Light	25
16/06	10.4	5.6	8.3	6	Light	25
17/06	12.8	.1	5.3	10	Light	27
18/06	13.0	7.4	3.2	9	Light	24
19/06	14.0	tr	.4	12	Moderate	33
20/06	12.6	0	7.7	6	Light	17

As previously noted, the **actual** data set has mean temperature for all locations. I changed to maximum temperature for this example for consistency with the textbook.

[Textbook]

(a) Describe the type of data represented by daily total rainfall.

Alison is investigating daily maximum gust. She wants to select a sample of size 5 from the first 20 days in Hurn in June 1987. She uses the first two digits of the date as a sampling frame and generates five random numbers between 1 and 20.

b) State the type of sample selected by Alison.

c) Explain why Alison's process might not generate a sample of size 5.

a a **Continuous quantitative data.**

b b **Simple random sample.**

c c **Some of the data values are not available (n/a).**

Hurn

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Daily maximum temperature and rainfall: 09:00–09:00. Sunshine, mean wind and maximum gust: 00:00–24:00.

June 1987	Max °C	Rain mm	Sun h	Wind kn	Beaufort	Gust kn
01/06	15.1	.6	4.5	7	Light	19
02/06	12.5	4.7	0	7	Light	22
03/06	13.8	tr	5.6	11	Moderate	25
04/06	15.5	5.3	7.8	7	Light	17
05/06	13.1	19	.5	10	Light	33
06/06	13.8	0	8.9	19	Fresh	46
07/06	13.2	tr	3.8	11	Moderate	27
08/06	12.9	1	1.7	9	Light	19
09/06	11.2	tr	5.4	6	Light	19
10/06	9.2	1.3	9.7	4	Light	n/a
11/06	12.6	0	12.5	6	Light	18
12/06	10.4	0	11.9	5	Light	n/a
13/06	9.6	0	8.6	5	Light	15
14/06	10.2	0	13.1	5	Light	18
15/06	9.2	3.7	7.1	4	Light	25
16/06	10.4	5.6	8.3	6	Light	25
17/06	12.8	.1	5.3	10	Light	27
18/06	13.0	7.4	3.2	9	Light	24
19/06	14.0	tr	.4	12	Moderate	33
20/06	12.6	0	7.7	6	Light	17

[Textbook] Calculate:

a) The mean daily maximum temperature for the first five days of June in Hurn in 1987.

b) The median daily total rainfall for the week of 14th June to 20th June inclusive.

c) The median daily total rainfall for the same week in Perth was 19.00mm. Karl states that more southerly countries experience higher rainfall during June. State with a reason whether your answer to part (b) supports this statement.

a

a

$70.0 \div 5 = 14.0°C(1dp)$

b

b

Values in ascending order:

0, 0, tr, 0.1, 3.7, 5.6, 7.4.

$\therefore$ Median is 0.1mm.

c

Perth is in Australia, south of the UK, and its median rainfall was higher. However, this very small sample from one location in each country does not provide enough evidence to support Karl?s statement.

Questions 3 and 4 use the following extracts from the large data set.

Leeming · © Crown Copyright Met Office 2015

Date	Mean temp. (°C)	Total rain (mm)	Sunshine (h)	Mean wind (kn)
01/06/15	8.9	10	5.1	15
02/06/15	10.7	tr	8.9	17
03/06/15	12.0	0	10.0	8
04/06/15	11.7	0	12.8	7
05/06/15	15.0	0	8.9	9
06/06/15	11.6	tr	5.4	17
07/06/15	12.6	0	13.9	10
08/06/15	9.4	0	9.7	7
09/06/15	9.7	0	12.1	5
10/06/15	11.0	0	14.6	4

Heathrow · © Crown Copyright Met Office 2015

Date	Mean temp. (°C)	Total rain (mm)	Sunshine (h)	Mean wind (kn)
01/06/15	12.1	.6	4.1	15
02/06/15	15.4	tr	1.6	18
03/06/15	15.8	0	9.1	9
04/06/15	16.1	.8	14.4	6
05/06/15	19.6	tr	5.3	9
06/06/15	14.5	0	12.3	12
07/06/15	14.0	0	13.1	5
08/06/15	14.0	tr	6.4	7
09/06/15	11.4	0	2.5	10
10/06/15	14.3	0	7.2	10

Exercise 1E – Pearson Statistics Y1 page 13-15 — Solution 37

3. Use the sunshine data for the first ten days of June 2015.

(a) Find the mean daily total sunshine for (i) Leeming and (ii) Heathrow.

(b) Find the range for (i) Leeming and (ii) Heathrow.

(c) Supraj says the further north you are, the fewer hours of sunshine. Do your answers support this? Give reasons.

3a(i) Leeming: $101.4/10 = 10.1$ h (1 d.p.).

3a(ii) Heathrow: $76/10 = 7.6$ h.

3b(i) Leeming: $14.6 - 5.1 = 9.5$ h.

3b(ii) Heathrow: $14.4 - 1.6 = 12.8$ h.

3c

Leeming is further north and has the higher mean sunshine. The data do not support the conclusion. The range does not affect this comparison.

4. Calculate the mean daily total rainfall at Heathrow for the first ten days of June 2015. Explain how you treated 2, 5 and 8 June.

4

Each "tr" is a trace below 0.05 mm. Treat a trace as zero because it rounds to 0.0 mm to one decimal place.

$\frac{0.6+0\times3+0\times5+0.8}{10} = \frac{1.4}{10} = 0.14$ mm.

Exercise 1E – Pearson Statistics Y1 page 13-15 — Solution 38

5. Dominic investigates how average monthly temperature changed during summer 2015 in Jacksonville. He averages temperatures for two days each month and compares the means.

(a) Give one reason two days a month might be (i) a good sample size, (ii) a poor sample size.

(b) He takes the first and last day of each month. Why might this be unrepresentative?

(c) Suggest a better way to choose days.

a(i)

Sampling from multiple months provides more data.

a(ii)

Two days out of a month of 30 or 31 days is a small sample.

b

The last day of one month and the first day of the next are consecutive and may share the same weather pattern. The selection is not random.

c

Number the days and take a simple random sample.

7. Brian calculates mean cloud coverage in Leeming in September 1987 as 9.3 oktas. Explain why this is incorrect.

7
One okta is one eighth of the sky. Cloud cover cannot exceed 8 oktas, so a mean of 9.3 is impossible.

8. The large data set contains 184 consecutive days from 1987. Marie investigates daily mean windspeeds in Camborne.

(a) Describe how to take a systematic sample of 30 days. (3 marks)

(b) Why would this sample not necessarily give 30 data points for her investigation? (1 mark)

8a
Put the days in date order. Since $184/30 \approx 6$, choose a random starting day from 1 to 6. Take every sixth day thereafter until 30 days have been selected.

8b
Some windspeeds may be missing from the data set.